

Sundareswari Thiyagarajan

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- » **Research Interests:** Machine Learning, Deep Learning, NLP, Computer Vision, Human Computer Interaction.

»» EDUCATION

B.E (Computer Science and Engineering)

August 2019 - June 2023

- » Anna University , (BIT Campus), Tiruchirappalli, Tamil Nadu, India.
- » CGPA : 8.34 / 10 (or) GPA : 3.75 / 4.5

High School

June 2017 - March 2019

- » Modern Matric Hr. Sec School, Tamil Nadu, India.
- » Marks : 81.67 %

Grade 10

June 2016 - March 2017

- » Modern Matric Hr. Sec School, Tamil Nadu, India.
- » Marks : 98.4 %

»» RESEARCH EXPERIENCE

- » **A Multi-Model Approach- Stress Detection using Physiological Signals with LSTM and XGBoost.** *IECE Transactions on Swarm and Evolutionary Learning 2025* (Under Review).
- » **The Multiple Approaches for Drug-Drug Interaction Extraction using Machine learning and transformer-based Model.** *Korea Science 2024* (Under Review)

»» INTERSHIPS

- » AI to detect Criminal Cars using License Plate under the Supervision of The Korean Academy. June 2024 - Aug 2024
- » Artificial Intelligence - Object Detection by Raksa Solution 10 Oct 2023 - 05 April 2024

»» PROJECT EXPERIENCE

- » **Stock Market Prediction using Sentimental Analysis and LSTM.** Feb 2023 - May 2023
Using NLP and machine learning techniques to classify each piece of text as positive, negative, or neutral sentiment. LSTM and polarity score are used to predict the stock prices.
Technologies used: Natural language processing, web scraping.
- » **Viable Embryo Detection in Banana Seed using Arduino .** Mar 2022 - June 2022
Detecting banana seeds using machine learning involves a combination of computer vision and machine learning techniques. It is a handheld device to detect the embryo and endosperm in banana seeds using Aurdino.
Technologies used: Image processing, Arduino, Convolutional Neural Networks (CNN)

- July 2022 - Dec 2022**

» **The Parkinson's Disease Detection using Machine Learning.**
 Detecting Parkinsons disease using Spiral test , where the input images are trained with Random Forest Classifier. To implement Parkinson's detector deep learning and Convolutional Neural Networks (CNNs) are used.
 Technologies used: OpenCV, Computer Vision, (CNN).
- Aug 2022 - Nov 2022**

» **Online Food Ordering Website.**
 Designing a Responsive User interface for ordering food using PHP, HTML, CSS. Datasets are stored in the PhpMyAdmin SQL database as well as in the Admin Panel. To create a System Layout Architecture, UI and database will be connected using PHP.
 Technologies used: PHP, HTML, CSS and XAMPP control panel

»»» COMPUTER SKILLS

- » **Technical Skills:** Python, C, Java, MySQL, HTML, CSS.
- » **Software:** Arduino IDE, Matlab, Visual Studio, Microsoft office suite, Adobe Illustrator, Adobe Xd, Adobe Photoshop.

»»» VOLUNTEER ACTIVITIES

- Sep 2019 - Mar 2022**

» NCC Cadet (C - Certificate Holder)
- 2022**

» Participated in Combined Annual Training Camp (CATC)
- June 2016 - Feb 2019**

» Participated in Shot put and Throw ball competition (District level)

Activities: Physical fitness, drill, weapon handling, social service, Map Reading, Obstacle and adventure sports.

Personality Development: The NCC experience helps in developing self confidence, communication skills, and overall personality growth.

»»» AWARDS AND CERTIFICATIONS

- Oct 2019**

» Texas Instruments Online contest participation in the IICDC-2019-LIC.
- May 2022**

» React JS Workshop conducted by Madras Institute of Technology, Anna University, Chennai.
- July - Dec 2022**

» Handheld portable device for viable embryo detection in banana breeding program funded by CSRC under Research Support Scheme "Student Innovative Project".
- Sep 2019 - Mar 2022**

» NCC 'C' Certificate provided by Ministry of Defence, Government of India.
- June - Aug 2024**

» Internship Certificate for AI to detect Criminal Cars with License Plate.
- July - Oct 2024**

» Hands-on approach to AI for real-world applications by IIT Kharagpur AI4ICPS, Hub Foundation and TCS ion